

A low-voltage mixed-mode CMOS/SOI integrated circuit for 13.56 MHz RFID applications

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Abstract

A low-voltage mixed-mode CMOS/SOI integrated circuit for 13.56 MHz RFID applications is presented. When connected to an external inductive loop antenna and put in a reader magnetic field, the chip is remote powered. Nominal supply voltage lowering down to 1.2 V (digital part) and 1.5 V (analog part) was achieved using a 0.25 μm partially-depleted CMOS/SOI technology. Device level simulations were done with the LETISOI model. Measurements proved the chip functionality and the suitability of the technology for lower voltages. To our knowledge, it is the first mixed-mode CMOS/SOI chip dedicated to RFID.

Introduction

RF identification (RFID) systems are currently under tremendous growth due to their wide application field (e.g. access control, luggage tracking ...). To meet the market requirements, such systems must exhibit features like low cost, long read range and high data rate, which require small and low-voltage/low-power integrated circuits [1]. In order to investigate the suitability of CMOS/SOI for these applications, we have designed and fabricated an ASIC for a long distance multi-tag RFID system.

Chip block diagram

The integrated circuit is composed (see fig. 1) of an RF front-end (tuning capacitor, voltage rectifier, voltage limiter, modulator, demodulator and clock recovery), an analog front-end (voltage reference, regulator, Power-On-Reset) and a digital part. Thanks to the voltage rectifier and voltage limiter, the chip drains power from the LC tank, regulates the antenna voltage to 2.5 V peak-to-peak, provides a 1.5 V supply voltage to the other analog blocks, a regulated 1.2 V supply voltage and a 1.7 MHz clock to the digital block.

Digital block design

Several deterministic anti-collision algorithms, LETI patented, are implemented to ensure the tag identification and the multi-tag functionality. For the tag to reader transmission, a 212 kHz sub-carrier is DQPSK modulated and transmitted to the front-end where it drives the load modulator. The data rate can be set to 212, 106 or 53 kbits/s. For the reader to tag

transmission, the digital block decodes a PPM modulated signal coming out from the amplitude demodulator. The data rate can be adjusted according to radio emission regulation from 2 kbits/s to 22 kbits/s.

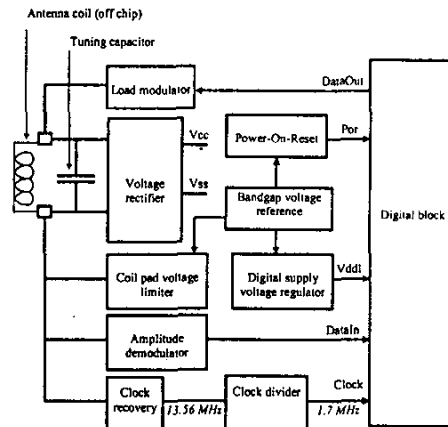


Fig. 1: Chip block diagram

The digital block was developed using a standard VHDL design flow, based on a digital library from LETI [2]. All cells use floating-body MOSFETs. The digital core (test pads excluded) is evaluated to about 12k gates.

Analog front-end design

In the design of these low voltage analog blocks, the main advantage of SOI over bulk CMOS came from the use of a gate-to-body connection (GBC also referred as DTMOS [3]). Although DTMOS is compatible with bulk process, its use is much more easier in SOI circuits. As an example, Fig. 2 shows the schematic of the voltage rectifier. Two floating-body NMOS switches are used to prevent the antenna pad voltage from reaching too negative values. Two body-contact GBC NMOS transistors are used as diodes. The gate-body connection of these MOSFETs lowers their threshold voltage and hence reduces the rectifier voltage drop, which is a key parameter for system performance.

All the simulations were done with the ELDO simulator and the LETISOI model [4].

Technology

The circuit was fabricated using a 0.25 μm partially-depleted CMOS/SOI technology developed at LETI on 200 mm SOITEC wafers [5]. This technology features floating body, body-tie and body-contact transistors of both N and P type, along with poly-diffusion capacitors and diffusion resistors. The DC threshold voltage is 0,6 V for N-type and -0,7 V for P-type transistors.

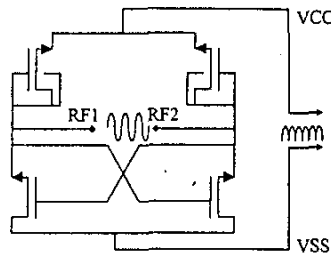


Fig. 2: Low voltage-drop rectifier using GBC diodes

Layout

Fig.3 shows a chip microphotograph including wire bonds. The total size is about 1.5 mm x 1.5 mm. Two pads are used to connect the chip to the antenna. The additional ones are used to test the digital part with a standard power supply or to probe the analog-to-digital interface signals.

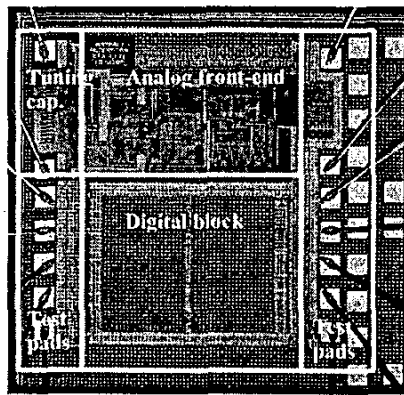


Fig. 3: Chip microphotograph

Experimental results

Half-duplex data transmission tests with a reader prove that, when coupled to a credit card size antenna, the circuit is fully functional for field intensity ranging from 0.5 to 2.5 A/m rms. To illustrate the remote power functionality of the chip, Fig. 4 shows the digital supply voltage versus field intensity. The digital block is functional for a supply voltage ranging from 1.8 V down to 0.6 V (external supply).

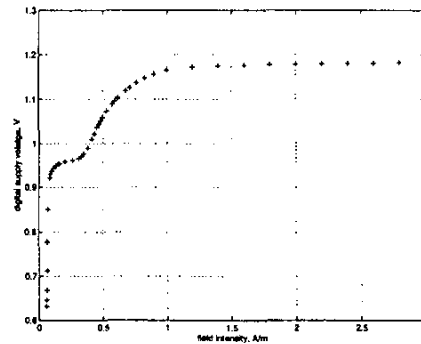


Fig. 4: Measured digital supply voltage vs field intensity

Conclusion

A 0.25 μm CMOS/SOI mixed-mode ASIC has been successfully designed and fabricated. A good agreement is found between measurements and ELDO simulations with LETISOI model. Taking advantage from some SOI unique features allowed us to achieve low-voltage operation for both the analog and digital parts.

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